

25CY206 STATISTICAL FOUNDATION FOR COMPUTING AND DATA SCIENCE

Hours Per Week :

L	T	P	SL	C
3	0	2	3	4

PREREQUISITE KNOWLEDGE: Basic knowledge in Mathematics and Statistics.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive introduction to probability and statistical methods with applications in computing. It covers fundamental probability concepts, probability distributions, statistical inference, estimation techniques, and regression analysis. The course also introduces advanced statistical methods such as Principal Component Analysis (PCA) and Time Series Analysis. Students will develop a strong foundation in probabilistic modelling and statistical reasoning, which are essential for data analysis, machine learning, and decision-making in computing and engineering.

MODULE-1

UNIT-1

18L+0T+12P+18SL=48 Hours

FUNDAMENTALS OF PROBABILITY:

Introduction to probability and its applications in computing, Probability axioms, Conditional probability and independence, Bayes' theorem and its applications. Random variables: Distribution function, discrete random variable, probability mass function, continuous random variable, probability density function, Expectation, variance, and moments.

UNIT-2

PROBABILITY DISTRIBUTIONS:

Discrete probability distributions: Uniform, Bernoulli, Binomial, Poisson, and Geometric. Continuous probability distributions: Uniform, Normal, Exponential, Weibull, Cauchy, Logistic. Central Limit Theorem and its significance, Joint, marginal, and conditional distributions. Covariance and correlation.

PRACTICES:

- Implement probability axioms, conditional probability, and Bayes' theorem using programming (Python/R).
- Understand and visualize different probability distributions using real-world data.
- Compute statistical measures for discrete and continuous random variables.
- Analyze relationships between two random variables using joint, marginal, and conditional distributions.
- Apply t-test to evaluate the logical consistency of probabilistic models in data science, such as reasoning through the validity of Bayes' theorem for predictive algorithms in computing applications.

MODULE-2

UNIT-1

27L+0T+18P+27SL=72 Hours

ESTIMATION AND STATISTICAL INFERENCE:

Sampling techniques and sampling distributions, Point estimation and properties of estimators. Maximum Likelihood Estimation (MLE). Confidence intervals for population parameters. Hypothesis testing: Null and alternative hypotheses, level of significance, p-value, critical value, Type I and Type II errors, Single sample and two sample tests for mean, proportion, variance and standard deviation.



- <https://i.ytimg.com/vi/FmWz4deqomaxresdefault.jpg>

UNIT-2**REGRESSION ANALYSIS:**

Simple linear model and assumptions, least-squares estimators and their properties, ML estimation of the parameters, statistical inference and prediction with the simple regression model, tests for normality, lack-of-fit testing, transformations on Y or X (Box-Cox, square root, log etc), Logistic regression.

UNIT-3**ADVANCED STATISTICAL METHODS:**

Principal Component Analysis (PCA) for Dimensionality Reduction, Introduction to Time Series Analysis and Forecasting.

PRACTICES:

- Perform point estimation, confidence intervals, and hypothesis testing on real-world datasets.
- Apply regression techniques for predictive modelling in computing applications.
- Reduce the dimensionality of large datasets while retaining essential features.
- Analyze and predict trends using time-series data in computing and finance.
- Integrate statistical methods with machine learning for practical applications

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1.	Understand the fundamental concepts of probability, random variables, and probability distributions and apply them in computing applications.	Understand,	1, 2, 5
2.	Analyze different probability distributions, their properties, and significance in real-world applications, including computing and machine learning.	Apply	1, 2, 4
3.	Perform statistical inference, estimation, and hypothesis testing using different techniques and interpret the results effectively.	Evaluate	2, 3, 5
4.	Apply regression analysis techniques, including linear and logistic regression, and interpret results using statistical inference methods.	Apply	2, 4, 6
5.	Implement advanced statistical methods such as PCA and Time Series Analysis for data-driven decision-making in computing.	Analyze, Create	3, 5, 11

Sustainable Development Goals (SDGs)	Indian Knowledge System (IKS)
SDG 3: Good Health and Well-being	Yes
SDG 4: Quality Education	
SDG 5: Gender Equality (indirectly)	
SDG 8: Decent Work and Economic Growth	
SDG 9: Industry, Innovation, and Infrastructure	
SDG 17: Partnerships for the Goals	

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXT BOOKS:

1. Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press, 2019.
2. Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage Learning, 2015.

REFERENCE BOOKS:

1. Morris H. DeGroot, Mark J. Schervish, Probability and Statistics, 4th Edition, Pearson, 2011.
2. Casella, George, Roger L. Berger, Statistical Inference, 2nd Edition, Cengage Learning, 2002.
3. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.